

## Task: Algebra in class revision for Major Test 1

1. Factorise the following expression by taking out the highest common factor: -----

$$30x - x^2$$

2. Factorise the following expression by taking out the highest common factor: -----

$$15pqr - 4pst$$

3. Simplify  $\frac{14y}{16} + \frac{12y}{16}$  -----

4. Simplify the following: -----

$$\frac{4x}{3} - \frac{10x}{3}$$

5. Simplify the following:  $\frac{3x}{5} + \frac{4x}{7}$  -----

6. Simplify  $\frac{3x}{11} - \frac{3x-4}{55}$  . -----

7. Simplify the expression: -----

$$\frac{a}{7} \times \frac{a}{5}$$

8. Simplify  $\frac{3y}{11} \times \frac{5y}{9}$ .

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9. Simplify the expression:

$$\frac{k}{4} \div \frac{5}{m}$$

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10. Simplify the following:  $\frac{9x}{7} \div \frac{5}{12y}$

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11. Expand the following using binomial expansion:  
 $(x + 10)(x + 8)$

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12. Expand and simplify the following:  
 $(5w + 7)(4w + 3)$

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13. Expand the following and simplify:  $6u(8u - 2v)$

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14. Expand the following perfect square:  $(m - 4)^2$

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15. Expand and simplify the following:  
 $2x(2x + 4) + x(x + 4)$

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## Task: Algebra in class revision for Major Test 1

1. Factorise the following expression by taking out the highest common factor: -----

$$30x - x^2$$

Find the highest common factor (HCF) of the two terms and use  $AB - AC = A(B - C)$  to factorise.

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Since the highest numerical factor common to both terms is 1, the common algebraic component will be the HCF.

$$30x - x^2 = x(30 - x)$$

2. Factorise the following expression by taking out the highest common factor: -----

$$15pqr - 4pst$$

Find the highest common factor (HCF) of the two terms and use  $AB - AC = A(B - C)$  to factorise.

-----

Since the highest numerical factor common to both terms is 1, the common algebraic component will be the HCF.

$$15pqr - 4pst = p(15qr - 4st)$$

3. Simplify  $\frac{14y}{16} + \frac{12y}{16}$  -----

Since the denominators are the same, we can just add the terms in the numerator.

$$\frac{14y}{16} + \frac{12y}{16} = \frac{14y + 12y}{16}$$

Evaluate the numerator.

$$\frac{14y}{16} + \frac{12y}{16} = \frac{26y}{16}$$

Cancel out the common factors in the numerator and denominator.

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26y and 16 have a highest common factor of 2.

$$\frac{14y}{16} + \frac{12y}{16} = \frac{13y}{8}$$

4. Simplify the following: -----

$$\frac{4x}{3} - \frac{10x}{3}$$

Since both fractions have a common denominator of 3, we can express them as a single fraction.

$$\frac{4x}{3} - \frac{10x}{3} = \frac{4x - 10x}{3}$$

Simplify by collecting like terms on the numerator.

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As they are both terms in  $x$ ,  $4x - 10x$  can be simplified to  $-6x$ .

$$\frac{4x}{3} - \frac{10x}{3} = -\frac{6x}{3}$$

Simplify the fraction by dividing the numerator and denominator by their highest common factor.

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Divide each term in the numerator and denominator by 3.

$$\frac{4x}{3} - \frac{10x}{3} = -2x$$

5. Simplify the following:  $\frac{3x}{5} + \frac{4x}{7}$  -----

35 is the lowest common multiple of 5 and 7, so rewrite both fractions with a denominator of 35.

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Multiply the first term by  $\frac{7}{7}$  and the second term by  $\frac{5}{5}$  to make the denominators of both fractions  $5 \times 7$ .

$$\frac{3x}{5} + \frac{4x}{7} = \frac{3x \times 7}{5 \times 7} + \frac{4x \times 5}{7 \times 5}$$

Evaluate the products in each numerator and denominator.

$$\frac{3x}{5} + \frac{4x}{7} = \frac{21x}{35} + \frac{20x}{35}$$

Since we now have a common denominator, express as a single fraction by combining numerators.

$$\frac{3x}{5} + \frac{4x}{7} = \frac{21x + 20x}{35}$$

Simplify the like terms in the numerator.

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Since the numerator has two terms containing a single power of  $x$ ,  $21x + 20x$  can be simplified.

$$\frac{3x}{5} + \frac{4x}{7} = \frac{41}{35}x$$

6. Simplify  $\frac{3x}{11} - \frac{3x-4}{55}$ .

Rewrite the fractions in terms of their lowest common denominator.

Since 55 is a multiple of 11, the lowest common denominator is 55.

$$\frac{3x}{11} - \frac{3x-4}{55} = \frac{15x}{55} - \frac{3x-4}{55}$$

Since the denominators are the same, we can just subtract the numerators.

$$\frac{3x}{11} - \frac{3x-4}{55} = \frac{15x - (3x-4)}{55}$$

Remove the brackets in  $-(3x-4)$  by multiplying each term by  $-1$ .

$$\frac{3x}{11} - \frac{3x-4}{55} = \frac{15x - 3x + 4}{55}$$

Collect like terms in the numerator.

Since the numerator has two terms containing a single power of  $x$ ,  $15x - 3x$  can be simplified.

$$\frac{3x}{11} - \frac{3x-4}{55} = \frac{12x + 4}{55}$$

7. Simplify the expression:

$$\frac{a}{7} \times \frac{a}{5}$$

Multiply numerator by numerator and denominator by denominator to express as a single fraction

Numerator becomes  $a \times a = a^2$

Denominator becomes  $7 \times 5 = 35$

$$= \frac{a^2}{35}$$

8. Simplify  $\frac{3y}{11} \times \frac{5y}{9}$ .

Cancel out common factors across the numerators and denominators.

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The highest common factor of  $3y$  and  $9$  is  $3$ , while the highest common factor of  $5y$  and  $11$  is  $1$ .

$$= \frac{y}{11} \times \frac{5y}{3}$$

Multiply numerator by numerator and denominator by denominator.

$$= \frac{y \times 5y}{11 \times 3}$$

Evaluate the products in the numerator and denominator.

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Remember that to multiply two terms of the same base, you have to add their indices ie

$$y^p \times y^q = y^{p+q}.$$

$$= \frac{5y^2}{33}$$

9. Simplify the expression:

$$\frac{k}{4} \div \frac{5}{m}$$

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Dividing by a fraction is equal to multiplying by the reciprocal of the fraction.

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The reciprocal of  $\frac{5}{m}$  can be found by swapping its numerator and denominator.

$$= \frac{k}{4} \times \frac{m}{5}$$

We can write this as a single fraction by multiplying numerator by numerator and denominator by denominator.

$$= \frac{km}{20}$$

10. Simplify the following:  $\frac{9x}{7} \div \frac{5}{12y}$

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Let's first rewrite the division as a multiplication.

Dividing by a fraction is the same as multiplying by its reciprocal.

What is the reciprocal of  $\frac{5}{12y}$  ?

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The reciprocal of a fraction is found by switching the numerator and denominator around.

$$= \frac{9x}{7} \times \frac{12y}{5}$$

Multiply numerator by numerator and denominator by denominator to express as a single fraction.

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The numerator becomes

$$(-9x) \times 12y = -108xy$$

The denominator becomes  $7 \times 5 = 35$

$$= \frac{108xy}{35}$$

11. Expand the following using binomial expansion:  
 $(x + 10)(x + 8)$  -----

Use the distributive law to expand the brackets  
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Multiply  $x$  with each term in  $x + 8$  and 10  
with each term in  $x + 8$ .

$$(x + 10)(x + 8) = (x)^2 + 8x + 10x + 80$$

Collect like terms and simplify  
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$$8x + 10x = 18x$$

$$(x + 10)(x + 8) = x^2 + 18x + 80$$

12. Expand and simplify the following:  
 $(5w + 7)(4w + 3)$  -----

To expand  $(5w + 7)(4w + 3)$ , think of  $(4w + 3)$  as a single quantity.

We can then apply the distributive law  $A(B + C) = AB + AC$  by multiplying  $(4w + 3)$  by both  $5w$  and  $7$ .

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We can rewrite the distributive law as  $(x + a)(x + b) = x(x + b) + a(x + b)$ .

We can use this law to multiply  $5w$  with  $(4w + 3)$  and  $7$  with  $(4w + 3)$ .

$$(5w + 7)(4w + 3) = 5w(4w + 3) + 7(4w + 3)$$

We want to expand both of  $5w(4w + 3)$  and  $7(4w + 3)$  using the distributive law  $A(B + C) = AB + AC$ .

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Multiply  $5w$  by each term in  $(4w + 3)$  and  $7$  by each term in  $(4w + 3)$

$$(5w + 7)(4w + 3) = 20w^2 + 15w + 28w + 21$$

Collect like terms

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$15w$  and  $28w$  are like terms. We can combine them by adding the coefficients.

$$(5w + 7)(4w + 3) = 20w^2 + 43w + 21$$

13. Expand the following and simplify:  $6u(8u - 2v)$  -----

Use the distributive law  $A(B + C) = AB + AC$   
to expand the brackets.

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Remember that  
*positive*  $\times$  *negative* = *negative* and  
*negative*  $\times$  *negative* = *positive*

$$= 48u^2 - 12uv$$

14. Expand the following perfect square:  $(m - 4)^2$  -----

Use perfect square expansion:

$$(A - B)^2 = A^2 - 2AB + B^2$$

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Here  $A = m$  and  $B = 4$ .

$$(m - 4)^2 = m^2 - 2 \times 4m + 4^2$$

Evaluate the product  $2 \times 4m$  and the square  $4^2$ .

$$(m - 4)^2 = m^2 - 8m + 16$$

15. Expand and simplify the following:

$$2x(2x + 4) + x(x + 4)$$

By the order of operations we want to first expand the multiplication of  $2x$  across the first set of brackets, and the multiplication of  $x$  across the second set of brackets.

We can use the distributive law

$$A(B + C) = AB + AC$$

To use the distributive law we multiply all terms in the first set of brackets by  $2x$  and all terms in the second set of brackets by  $x$ .

$$2x(2x + 4) + x(x + 4) =$$

$$2x \times 2x + 2x \times 4 + x^2 + x \times 4$$

By the order of operations we evaluate multiplication before addition.

Simplify  $2x \times 2x$  and  $2x \times 4$

$$2x(2x + 4) + x(x + 4) = 4x^2 + 8x + x^2 + 4x$$

Collect like terms.

Add the terms in  $x^2$ , and also the terms in  $x$ .

$$2x(2x + 4) + x(x + 4) = 5x^2 + 12x$$